ANDRÉ ROGÉRIO DA SILVA FILHO

Applied AI Engineer & Software Developer | Enterprise Integrations • Agentic Systems • Cloud Architecture

Sorocaba, SP, Brazil (Open to Remote / CET Overlap) • Spanish Citizenship in Progress (EU Work Eligibility) •
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SUMMARY

Applied AI engineer and software developer with an operational foundation established across corporate IT and industrial manufacturing. Build LLM-based agents, enterprise integrations (ERP to Copilot Studio), and telemetry pipelines in TypeScript and Python, emphasizing deterministic guardrails, verifiable behavior, and data privacy. Currently an ICT Resident Researcher in industrial digital twins with UniFacens, Softex, and the Ministry of Science, Technology and Innovation (MCTI). Based in Brazil (UTC-3), open to remote work with CET overlap.

TECHNICAL SKILLS

Languages & Runtimes: Python, TypeScript, JavaScript, SQL, Node.js, PowerShell

Software Architecture & Web: React, Next.js, Fastify, Express, WebSockets, REST APIs, Tailwind CSS

Applied AI & Agents: LLM Agents, Tool Calling, Google Gemini API, OpenAI API, Vision Models, RAG Architectures, Fuzzy Matching

Cloud, Data & Governance: Google Cloud (Cloud Run, Cloud SQL), Microsoft Purview, PostgreSQL (Row-Level Security), Docker, Redis, CI/CD, AppSec

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura)

07/2026 - Present

Software Development Consultant (Contractor) | São Paulo, Brazil (Remote) (10/2026 - Present)

- Architected Lifecycle OS (Onboarding OS), a multi-tenant employee admission platform in closed beta expanding to a second public concession, cutting document audit time from 30+ minutes down to 10 minutes per hire (66%+ reduction) while enforcing strict LGPD compliance and PostgreSQL Row-Level Security.
- Engineered custom TypeScript middleware bridging Sienge ERP with Microsoft Copilot Studio, deploying an in-memory fuzzy matching engine (Token Sort and Levenshtein distance) with an 80% similarity threshold to eliminate duplicate catalog materials across regional construction sites.
- Architected the CBI Operations Center on Google Cloud (Cloud Run, Cloud SQL), consolidating incident ticketing and fleet telemetry to reduce projected operational licensing costs compared to commercial enterprise ITSM suites.
- Designed and deployed 'Isa', an AI service agent using Google Gemini and structured tool calling for automated SLA classification, intent routing, and tier-1 ticket resolution across concession operations (pilot validated with 22 active users).
- Lead enterprise data governance using Microsoft Purview and facilitate corporate workshops to train end-users on responsible AI adoption, data compliance, and secure prompt workflows.

IT Analyst | São Paulo, Brazil (Remote) (07/2026 - 10/2026)

- Developed full-stack web applications and internal tools using TypeScript, React 19, Next.js, Node.js, and PostgreSQL.
- Built automated API integrations connecting operational platforms with Microsoft Graph and identity governance services.
- Created custom automation scripts in Python and TypeScript to eliminate manual data entry, resulting in promotion to Software Consultant.

Farmamed Drogarias

11/2025 - 02/2026

Support and Infrastructure Analyst | Sorocaba, Brazil

- Administered multi-site IT infrastructure and user access for a regional retail pharmacy network, ensuring continuous point-of-sale availability and endpoint security.
- Maintained endpoint compliance and identity governance using Microsoft Entra ID and Intune across branch locations.
- Automated diagnostic data extraction and log verification using Python scripts to accelerate incident resolution.

Wobben Windpower (ENERCON Group)

11/2023 - 01/2025

Systems & Support Intern | Sorocaba, Brazil

- Supported day-to-day corporate IT operations and user workstations for the Brazilian wind turbine manufacturing facility in alignment with ENERCON global guidelines.
- Participated in the global Microsoft Copilot Flight Crew pilot organized by ENERCON Global IT, evaluating generative AI enterprise features and practical adoption in corporate workflows.
- Maintained and executed PowerShell automation scripts to standardize OS deployments, ensuring full compliance with European headquarters security policies.
- Liaised directly in English with global enterprise consultants (Infosys) to coordinate user access provisioning and SAP permissions, managing corporate service tickets via ServiceNow under an official ENERCON domain account.

SELECTED APPLIED R&D & ENGINEERING INITIATIVES

CRIBUM (AuditRx) | Offline-First Pharmaceutical Safety Assistant Portfolio

Selected for Campus Mobile (Instituto Claro & Poli-USP) • AMD Developer Hackathon • Python, FastAPI, Vision, Graphs

- Architected an offline-first assistant to protect patients from adverse drug interactions and dosage errors by verifying prescriptions and packaging on mobile devices.
- Integrated multimodal vision to extract active pharmaceutical ingredients, decoupling probabilistic OCR from a deterministic rules engine built from official health datasets (ANVISA/FDA) to prevent medical hallucinations.
- Implemented sub-second local verification pipelines backed by a 36-test automated suite (22 PyTest unit tests and 14 Playwright E2E scenarios) to ensure clinical reliability without internet connectivity.

FundiVR | Industrial Metallurgy Digital Twin & Telemetry Simulator GitHub

MCTI / Softex / UniFacens Technology Residency • TypeScript, React, Zustand, Fastify, PostgreSQL, WebSockets

- Engineered a full-stack digital twin simulation for a 700°C+ industrial aluminum melting furnace, modeling operational states, safety compliance, and emergency shutdowns.
- Built a low-latency WebSocket telemetry pipeline and real-time evaluation engine for simulated furnace telemetry, receiving a 10/10 capstone defense evaluation.

Memory Lens | 3D Spatial Knowledge Engine & Assistive Memory Platform Portfolio

Applied AI Research advised by Prof. Dr. Fabrício Londero (UniFacens) • Targeting submission to ACDSA • Three.js, RAG

- Designed an external episodic memory platform for neurodivergent individuals, mapping unstructured notes into semantic 3D vector space using Three.js and WebGL shaders to enable non-linear visual exploration.
- Engineered a contextual RAG pipeline and semantic search using vector embeddings and Gemini, enabling non-linear concept retrieval across unstructured notes.

TECHNOLOGY RESIDENCY & APPLIED RESEARCH

ICT Resident Researcher (Digital Twins & Applied AI) 10/2026 - 04/2027 (Ongoing)

UniFacens | Softex | Ministry of Science, Technology and Innovation (MCTI) | ICT Residency 51

- Selected for a 6-month federal R&D research fellowship following Phase 1 graduation, engineering an enterprise industrial Proof of Concept (PoC) in digital twins, computer vision, and systems integration.

Advanced Technical Specialization (Digital Twins, XR, 5G & AI) 02/2026 - 09/2026

UniFacens | Softex | MCTI do Futuro Federal Program

- Completed 360-hour intensive specialization ranked 8th nationwide out of 300+ candidates with a 4.00 / 4.0 GPA (9.91 / 10.0 average) and 10 maximum scores, leading technical development across 3 capstone projects.

EDUCATION

Higher Technology Degree in Systems Analysis and Development (TADS) 11/2023 - 07/2026

Centro Universitário SENAC, Brazil | Coursework completed (diploma pending ENADE) | GPA approx. 3.38 / 4.0

CERTIFICATIONS & LANGUAGES

Certifications: Certified AppSec Practitioner (CAP, The Security Academy / Instituto de Defesa Cibernética) | Google Cloud Generative AI Leader

Professional Training: Cisco CCNA 1 (Networking Basics) | Microsoft Azure Well-Architected Framework (Fast Lane)

Languages: English: Fluent (C1) | Spanish: Intermediate / Technical | Portuguese: Native

Community: Active volunteer host at Clube Poliglota Sorocaba, facilitating conversational language immersion sessions.